

Chapter 4: In-Class Exercises

4.2

Problem: Chinh is presented an annuity that is to pay \$4,000 at the end of each six months for six years. Using compound interest at a nominal monthly interest rate of 3%, what is the value of Chinh's reward at the time of the announcement (six months before the first \$4,000 payment)?

$$\frac{i^{(12)}}{12} = .0025$$

12 pmts in 6 month increments

$$4000 a_{\overline{12}|i} = 4000 \left(\frac{a_{\overline{72}|.0025}}{s_{\overline{6}|.0025}} \right)$$

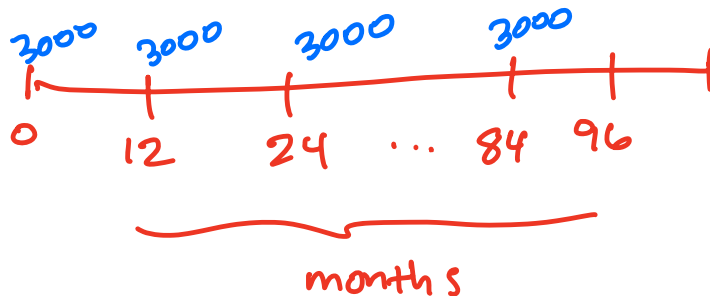
↑
6 month effective rate

$$= 43604.47$$

4.2

Problem: Josh makes annual beginning-of-year contributions of \$3,000 to a retirement account. The account has a nominal interest rate of 4.8% convertible monthly. Find Josh's balance just before his ninth contribution by finding the annual effective interest rate and then check your answer using (4.2.8).

$$\frac{i^{(12)}}{12} = .004$$



$$3000 \ddot{s}_{\overline{9}|i} = 3000(1.004)^{96} \ddot{a}_{\overline{9}|i} = 3000(1.004)^{96} \left(\frac{a_{\overline{96}|.004}}{a_{\overline{12}|.004}} \right)$$

↑
annual effective rate

$$= 29953.30$$

4.2

Problem: A perpetuity paying \$1,650 at the end of each year is to be replaced by another paying X at the beginning of each two years. The present value of each perpetuity is \$31,250. Find X .



$$31,250 = \frac{1650}{i} \Rightarrow i = \frac{1650}{31250}$$

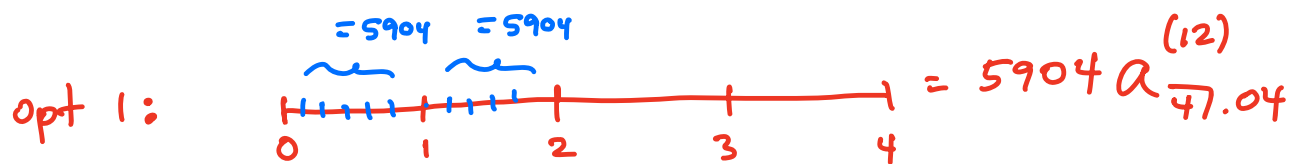
$$31250 = X \left(\frac{1}{i a_{\infty|i}} \right)$$

$$X = 3055.90$$

4.3

Problem: Mahpee was offered a scholarship with level monthly payments for four years. The first payment would be made on September 30, 2000, and each year the payments would total \$5,904. Mahpee, who was contemplating actuarial studies, wished to compare this scholarship to a fellowship offered by another college. The latter would pay \$5,800 on September 1st of years 2000, 2001, 2002, and 2003. Assume that the growth of money is governed by compound interest at an annual effective interest rate of 4%, and compare the values of the two grants on September 1, 2000.

Given: annual $i = 4\%$



$$= 5904 \left(\frac{1 - (1.04)^{-4}}{((1.04)^{1/12} - 1)12} \right) = 21821.02$$

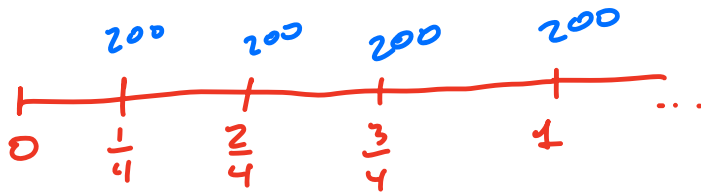


$$= 21895.53$$

4.3

Problem: Shirley invests \$200 at the end of each quarter for eight years in an account having an annual effective interest rate of 4%. Find Shirley's accumulated value at the time of her last deposit.

Given: annual $i = 4\%$.



$$\begin{aligned}
 & 800S_{\overline{31}.04}^{(4)} \\
 &= 800 \left(\frac{1 - (1.04)^{-8}}{((1.04)^{1/4} - 1)^4} \right) (1.04)^8
 \end{aligned}$$

$$= 7481.05$$

4.3

Problem: Find the numerical values of the four annuity symbols in inequality (4.3.14) when $n = 48$, $m = 12$, and $i = 5\%$.

$$\ddot{a}_{\overline{n}|i} = \frac{1 - 1.05^{-48}}{.05} (1.05) = 18.781015$$

$$\ddot{a}_{\overline{48}|.05}^{(12)} = \frac{1 - 1.05^{-48}}{.05} \left(\frac{.05}{d^{(12)}} \right) = 18.563098$$

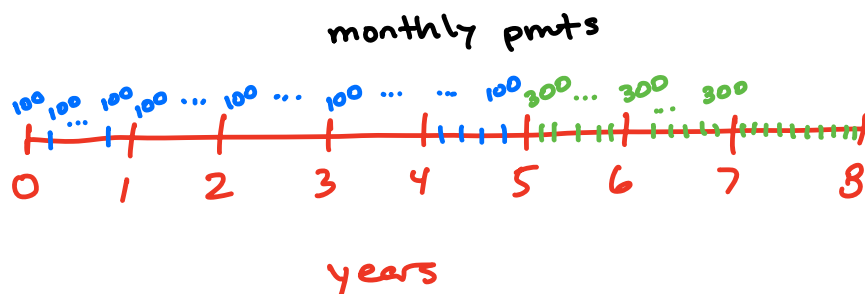
$$a_{\overline{48}|.05}^{(12)} = \frac{1 - 1.05^{-48}}{.05} \left(\frac{.05}{i^{(12)}} \right) = 18.48779674$$

$$\ddot{a}_{\overline{n}|i} = \frac{1 - 1.05^{-48}}{.05} = 18.07715782$$

4.3

Problem: Pavol purchases a perpetuity-due. The perpetuity has monthly payments of \$100 during the first five years, and then these increase to \$300. Determine the purchase price if it is based on an annual effective discount rate of 3.6%.

Given: annual discount $d = .036 \Rightarrow i = \frac{.036}{.964}$

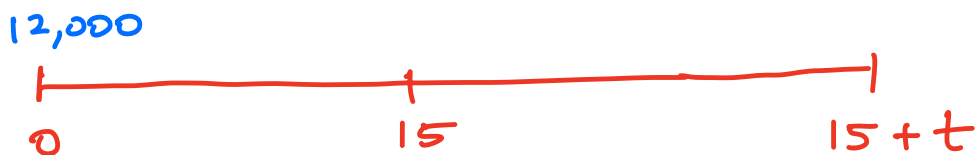


$$PV = \underbrace{1200 \ddot{a}_{\overline{60}|i}}_{\text{blue}} + \underbrace{(1-d)^5 \ddot{a}_{\overline{\infty}|i}}_{\text{green}} = 1200 a_{\overline{60}|i} \left(\frac{i}{d^{(12)}} \right) + .964^5 \left(\frac{1}{d^{(12)}} \right) \times 3600$$

$$= 87358.00$$

4.6

Problem: Martin deposits \$12,000 in a savings account with a discount rate of 10% convertible quarterly. He leaves his money in this account to accumulate for fifteen years, then moves it to a fund that is accumulating at 8% per annum convertible continuously. If, starting at time 15 when he invests in the new fund, money is withdrawn continuously at a rate of \$6,000 per annum, how long will Martin's money last?



$$12,000 \left(1 - \frac{1}{4} \right)^{-60} = 6000 \overline{a}_{\overline{t}|i} = 6000 \left(\frac{1 - e^{-.08t}}{.08} \right)$$

$$\Rightarrow t = 16.4 \quad \text{so to year}$$

$$31.4$$

4.6

Problem: Suppose that money grows according to simple discount at a rate of 3.6%. Find formulas for $\bar{a}_{\overline{n}|}$, $\bar{s}_{\overline{n}|}$, $\ddot{s}_{\overline{n}|}$.

$$a(t) = \frac{1}{1 - .036t}$$

$$v(t) = 1 - .036t$$

$$\begin{aligned}\bar{a}_{\overline{n}|} &= \int_0^n \frac{1}{1 - .036t} dt = \left. t - \frac{.036t^2}{2} \right|_0^n \\ &= n - \frac{.036n^2}{2}\end{aligned}$$

$$\bar{s}_{\overline{n}|} = \bar{a}_{\overline{n}|} \cdot a(n) = \frac{n - \frac{.036n^2}{2}}{1 - .036n}$$

4.6

Problem: An annuity is payable continuously for ten years. The rate of payment at time t is t^2 yearly. Assuming there is a constant force of interest $\delta = .04$, find the present value of the annuity.

$$\bar{a}_{\overline{10}|} = \int_0^{10} t^2 e^{-\delta t} dt$$

$$\hookrightarrow v(t) = e^{-\delta t}$$

$$u = t^2 \quad v = \frac{-1}{.04} e^{-.04t}$$

$$du = 2t dt \quad dv = e^{-.04t} dt$$

$$\frac{-t^2}{.04} e^{-.04t} + \underbrace{\frac{2}{.04} \int t e^{-.04t} dt}$$

$$u = t \quad v = \frac{1}{.04} e^{-.04t}$$

$$du = dt \quad dv = e^{-.04t} dt$$

$$\frac{-t}{.04} e^{-.04t} + \frac{1}{.04} \int e^{-.04t} dt$$

$$= \frac{-t}{.04} e^{-.04t} - \frac{1}{(.04)^2} e^{-.04t}$$

plugging this in

$$= \left(\frac{-t^2}{.04} e^{-.04t} + \frac{2}{.04} \left[\frac{-t}{.04} e^{-.04t} - \frac{1}{(.04)^2} e^{-.04t} \right] \right) \Big|_0^{10}$$

$$= 247.70$$